

Cosmology & Astronomical Instrumentation

lwkenuinely just buckets of light gng

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Prerequisite Knowledge

- Wave diffraction patterns
- Basic special relativity frames
- Some basic geometric optics
- Braincells, now magnified to 67420x

1 Introduction

Imagine you're a photon from the cosmic microwave background. You were released right at the moment the universe first became transparent, a mere 380,000 years after the Big Bang, back when the entire observable universe was a dense, glowing fog barely a thousandth of its current size. You set off in a straight line and just... kept going.

For the next 13.8 billion years, nothing gets in your way — but the space you're traveling *through* does not sit still. It keeps stretching, and stretching, and stretching, and every bit of that stretching drags your wavelength out right along with it, from a blistering near-infrared glow down to a barely-there microwave hum. You are not tired from moving fast; you are tired from the universe growing up around you.

Finally, after the longest trip anything has ever taken, you slam into a curved metal dish built by a small group of clever, carbon-based creatures on a rocky planet you've never heard of. Everything in the first half of this handout is about that final split second: the telescope that gathers you, the sensor that counts you, and the array of dishes working together that manages to resolve you against your billions of neighbors. Everything in the second half is about what your absurdly long journey can tell those creatures about how big, how old, and how fast their universe really is.

2 Practical Telescope Metrics and Optics

2.1 Light-Gathering Power and Focal Ratios

A telescope, at the end of the day, is just a very expensive bucket for catching light. Its most basic job is to catch as many photons as possible, and since photons rain down uniformly across the incoming wavefront, the number a telescope catches per second is proportional to its collecting *area*, not its diameter:

$$\boxed{\text{LGP} \propto D^2.}$$

This is the sleep paralysis demon for every observatory director: doubling a telescope's aperture doesn't double its light-gathering power, it *quadruples* it. Which is exactly why astronomers are willing to spend a billion dollars and a decade of construction just to gain a few extra meters of mirror — every meter buys back an enormous amount of faint, distant universe.

A second, independent property is the **focal ratio** (or “*f*-number”), the focal length divided by the aperture diameter:

$$\boxed{f/\# = \frac{f}{D}.}$$

This ratio, not the raw aperture size, controls how bright an *extended* image (a nebula, a galaxy, anything bigger than a point source) appears at the focal plane: a “fast” system (small $f/\#$) concentrates the same collected light onto a smaller image, while a “slow” system (large $f/\#$) spreads that same light out thinner, like the difference between a wide-open camera aperture and a pinhole. Point sources like stars don't care about any of this — their image brightness is set purely by the raw collecting area D^2 , since a star always gets squeezed into essentially the same tiny diffraction spot no matter what f -ratio you're using.

2.2 Diffraction-Limited Angular Resolution

This pisses off every telescope engineer equally: even a flawless mirror, ground with literally zero manufacturing error, still can't resolve arbitrarily fine detail. Light itself won't allow it. Any wave passing through a finite circular opening inevitably diffracts, spreading a perfect point source out into a small blurred disk (the **Airy disk**) surrounded by faint rings, no matter how perfectly the glass was polished.

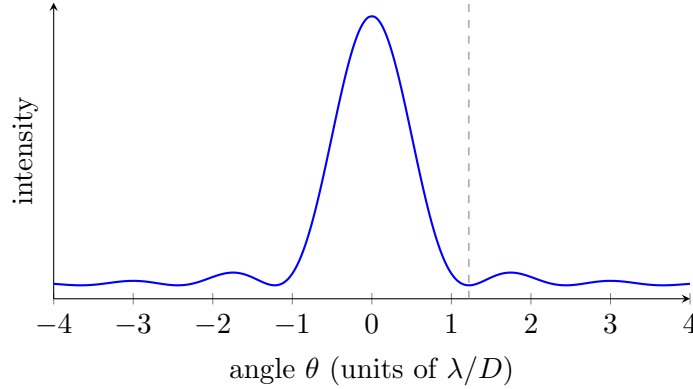


Figure 1: The Airy diffraction pattern from a circular aperture. The first dark minimum, conventionally taken as the resolution limit, sits at $\theta = 1.22\lambda/D$.

The exact intensity distribution, from treating the circular aperture as a diffracting wavefront (formally, the squared magnitude of the 2D Fourier transform of a uniformly illuminated disk), is

$$I(\theta) = I_0 \left[\frac{2J_1(x)}{x} \right]^2, \quad x = \frac{\pi D \theta}{\lambda},$$

where J_1 is the first-order Bessel function of the first kind. That function's first zero happens to occur at $x = 3.8317$, which is exactly where the numerical factor 1.22 in the Rayleigh criterion below comes from: solving $\pi D \theta / \lambda = 3.8317$ for θ gives $\theta = 1.22\lambda/D$. (Figure 1 actually plots the simpler sinc^2 function, matched to have its first zero at that same location, purely because it's easier to compute — the true Airy pattern's secondary rings are somewhat fainter than sinc^2 's, but the central peak and first null, which are all the Rayleigh criterion actually cares about, are essentially identical.)

Two point sources are considered just barely distinguishable (the **Rayleigh criterion**) the moment the peak of one source's Airy pattern falls right on top of the first dark minimum of the other's:

$$\theta \approx 1.22 \frac{\lambda}{D}.$$

And now you can see the trap closing: gathering more light wants a bigger D (Section 2.1), and so does resolving finer detail. Both roads lead to the same expensive place — a bigger mirror. But mirrors get exponentially harder and pricier to build past a certain size, which is exactly the pressure that eventually forces astronomers to stop building one enormous dish and start wiring several modest ones together instead (Section 3.3).

2.3 Plate Scale

So light's been gathered, it's been focused, and now there's a real, physical image sitting at the focal plane, ready to hit a detector. But that image is measured in millimeters, and the sky is

measured in degrees — we need a translator between the two.

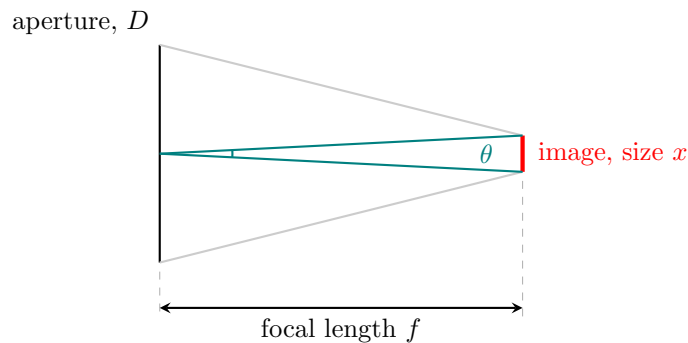


Figure 2: A small angle on the sky θ projects to a linear size x at the focal plane, related by the focal length f .

For a small angle θ (in radians), simple geometry gives $x = f\theta$. Converting θ from radians to arcseconds (using $206265''$ per radian) and rearranging for the conversion factor itself — the **plate scale** — gives

$$\text{plate scale} = \frac{206265''}{f \text{ (mm)}} \left[\frac{''}{\text{mm}} \right].$$

Multiplying a detector's physical pixel size (in mm) by the plate scale immediately gives that pixel's angular size on the sky — an essential number for deciding whether a given camera is well matched to a given telescope's diffraction limit (Section 2.1) at all.

3 Modern Sensors and Data Transformation

3.1 Charge-Coupled Devices (CCDs)

Once your photon actually lands, something has to count it. A CCD does this by converting incoming photons into stored electric charge, one tiny light-sensitive **pixel** at a time — it's basically a grid of several million microscopic buckets, each one just sitting there patiently collecting electrons. Three practical numbers decide how good a bucket it is:

- **Quantum efficiency (QE)**: the fraction of incoming photons that actually get converted into a stored electron (real CCDs regularly exceed 80–90%, dramatically better than the human eye or photographic film).
- **Full well capacity**: the maximum number of electrons a single pixel can hold before saturating; together with the noise floor, this sets the detector's dynamic range.
- **Dark current**: thermally generated electrons that sneak into a pixel's bucket even with no light falling on it at all — pure heat masquerading as signal, and utterly indistinguishable from the real thing once it's sitting there as charge. Which is precisely why sensitive CCDs are cooled, often to well below 0°C , to make the pixels sit still and stop manufacturing fake photons out of thermal noise.

3.2 Signal-to-Noise Ratio

Nothing in a real pixel is ever clean. The astronomical signal itself, the ambient sky glow, the dark current we just met, and the electronics' own read noise are all dumping electrons into the same

bucket, and every one of those sources jitters randomly from exposure to exposure. The trick to untangling them is that these noise sources are statistically independent, so it's their *variances* (not their raw magnitudes) that add up cleanly. Photon-counting processes (signal, sky, dark current) are Poisson-distributed, so each one's variance is simply its own mean count; read noise R is a fixed, exposure-independent contribution added in quadrature on top. Assembling all of that for a source of count rate $N_\star$ (electrons/sec), integrated for a time t over n_{pix} pixels with sky and dark count rates N_{sky} and N_{dark} , gives the workhorse formula every observer eventually memorizes:

$$\text{SNR} = \frac{N_\star t}{\sqrt{N_\star t + n_{pix} (N_{sky} t + N_{dark} t + R^2)}}.$$

In bright sources, the signal term $N_\star t$ dominates the noise too, so $\text{SNR} \rightarrow \sqrt{N_\star t}$ — meaning you only ever buy resolution at a steep discount, growing as the square root of exposure time. In faint sources swamped by sky background or read noise, the SNR instead grows linearly with t at first but eventually plateaus as those fixed backgrounds take over — which is exactly why long-exposure astrophotographers fight so hard, and drive so far from city lights, just to keep that background term small.

3.3 Two-Element Interferometry

Section 2.2 just handed us some bad news: a single dish's resolution is capped at $\theta \sim \lambda/D$, and for radio wavelengths (often centimeters to meters long), matching optical-quality resolution would need a single dish many kilometers wide. Nobody is building that. **Interferometry** is the clever way out: instead of one impossibly huge dish, wire together two (or more) perfectly ordinary, buildable dishes separated by a large **baseline** B , and let them cheat together.

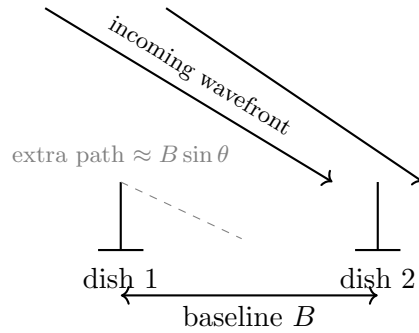


Figure 3: Two-element interferometry. Light from a source at angle θ off the array's axis reaches one dish before the other, with an extra path length $B \sin \theta$.

Light from a source at angle θ off the array's boresight reaches the two dishes with a path difference $\Delta = B \sin \theta \approx B\theta$ (small angle), corresponding to a phase difference $\phi = 2\pi\Delta/\lambda$ between the two signals once they're electronically combined. As θ sweeps (either because the source itself moves, or because Earth's rotation slowly turns the baseline relative to the sky), ϕ cycles through 2π every time θ changes by

$$\delta\theta_{fringe} = \frac{\lambda}{B},$$

producing alternating bright and dark **fringes** exactly like a double-slit experiment, with a fringe spacing set entirely by λ/B rather than by either dish's own aperture. A source's true angular position can therefore be pinned down to a precision comparable to this fringe spacing, and the

resulting angular resolution is set not by either dish's own diameter, but by the much larger separation between them:

$$\theta_{min} \sim \frac{\lambda}{B_{max}}.$$

(A real array uses many baselines simultaneously, at every antenna pair and every orientation Earth's rotation sweeps out, and combines all of them — a technique called **aperture synthesis** — to reconstruct a full image rather than just a single fringe pattern; the single-baseline picture here captures the resolution limit, but not the imaging itself.) An array of modest, individually cheap dishes spread across a continent (or, for space-based very long baseline interferometry, across an appreciable fraction of an AU) can therefore synthesize the angular resolution of a single dish far larger than anything actually buildable — at the cost of the array's total *collecting area* (and hence sensitivity to faint sources) remaining limited to the sum of the individual dishes' own areas.

4 The Kinematic Universe

4.1 The Cosmic Distance Ladder

Here's the problem with distances in astronomy: no single ruler is long enough. Whatever technique gets you started nearby runs out of range long before you reach the edge of the observable universe. So instead, astronomers climb a **ladder** of overlapping methods, each rung calibrated by standing on the one below it, and trusting that rung completely.

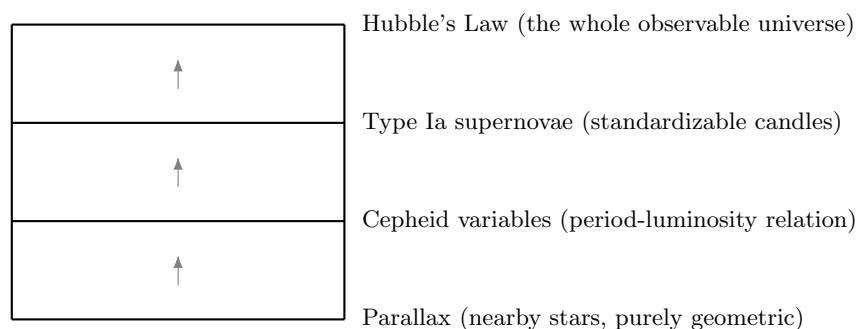


Figure 4: The cosmic distance ladder. Each rung's absolute calibration comes directly from the rung below.

- **Parallax:** purely geometric, requiring no assumptions at all — a nearby star's apparent position shifts against distant background stars as Earth orbits the Sun, with $d(pc) = 1/p''$. This anchors the entire ladder but only reaches a few thousand parsecs even with space-based astrometry.
- **Cepheid variables:** pulsating giant stars whose pulsation period is tightly correlated with their intrinsic luminosity (the **period-luminosity relation**). Once this relation is calibrated using nearby Cepheids with known parallax distances, observing a distant Cepheid's period alone reveals its true luminosity, and hence (via the inverse-square law) its distance — reaching out to tens of megaparsecs, far beyond the limits of parallax.
- **Type Ia supernovae:** the thermonuclear detonation of a white dwarf as it's pushed over the Chandrasekhar limit (Handout 5, Section 5.2) — since every such explosion involves almost exactly the same critical mass, they detonate with almost exactly the same peak luminosity,

making them “standardizable candles” bright enough to be seen across billions of light-years. Their absolute luminosity is calibrated using Cepheids observed in the same host galaxies.

Each rung is a cross-check on the one below it, which is precisely why the ladder is trusted — an error at any single rung would show up as a contradiction with independent measurements made using the next.

4.2 Hubble’s Law

Plot recession velocity (measured via redshift) against independently measured distance for a big enough sample of galaxies, and something almost eerie falls out: a strikingly clean straight line.

$$v = H_0 d.$$

This is a good local (low-redshift) approximation to cosmic expansion: every sufficiently large volume of the universe is expanding uniformly, so a galaxy twice as far away recedes, on average, twice as fast. That’s not a coincidence, and it’s not because every galaxy is fleeing some central explosion point — it’s the unmistakable signature of space itself stretching uniformly everywhere at once, with every galaxy just along for the ride.

4.3 Cosmological Redshift

At low redshift, it’s tempting (and mostly harmless) to read Hubble’s law as an ordinary Doppler shift, as if galaxies were just physically flying away from us through fixed, static space. But that picture “skgnakflgkjndsfgl”s itself at higher redshift, and the correct one is stranger: it’s space itself that stretches while the photon is mid-flight, dragging the photon’s wavelength along for the ride — exactly what happened to you, back in the introduction. This is captured by the universe’s **scale factor** $a(t)$ (normalized so $a_{now} = 1$), and the relationship is remarkably simple:

$$1 + z = \frac{a_{now}}{a_{then}} = \frac{\lambda_{obs}}{\lambda_{emit}}.$$

A photon emitted when the universe was half its present size ($a_{then} = 0.5$) is observed today at $z = 1$ — its wavelength has been stretched to exactly double what it started as, simply because every length scale in the universe, including that photon’s own wavelength, doubled right along with it. At low redshift, this reduces smoothly back to the familiar linear Hubble law ($cz \approx H_0 d$ for $z \ll 1$), but at high redshift, the two pictures (recession velocity vs. stretching space) diverge, and only the scale-factor description survives.

5 The Macroscopic Cosmological Model

5.1 The Cosmic Microwave Background

Remember you, the photon from the introduction? You’re not hypothetical. You’re the **cosmic microwave background (CMB)**, and you were released at the exact moment of **recombination**, when the universe first cooled (to $\sim 3000\text{ K}$) enough for electrons and protons to finally combine into neutral hydrogen, letting photons like you travel freely for the first time instead of being endlessly slapped around by free electrons.

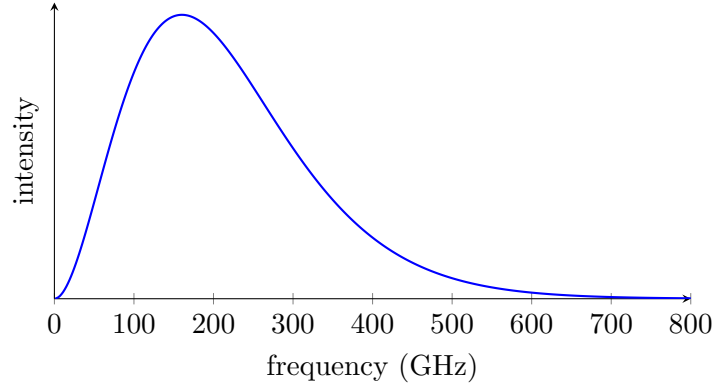


Figure 5: The CMB spectrum — the most perfect blackbody curve ever measured in nature, peaking at $\sim 2.7\text{ K}$ today, redshifted down from $\sim 3000\text{ K}$ at recombination by an expansion factor of roughly 1100.

Because the scale factor has grown by a factor of ~ 1100 since recombination, your temperature (which cools as $T \propto 1/a$, exactly like the wavelength stretching of Section 3.3) has dropped from a blistering $\sim 3000\text{ K}$ all the way down to a barely-there 2.725 K , spread uniformly across the whole sky. Your tiny anisotropies — fluctuations of only about 1 part in 10^5 — are a direct baby picture of the density fluctuations that would go on to seed every galaxy and galaxy cluster in today’s universe. Not bad, for something that spent 13.8 billion years just falling through empty space.

5.2 The Friedmann Equation

Zoom out from any single photon, and ask the biggest possible question: is the whole universe going to keep expanding forever, or is it going to slow down, stop, and fall back in on itself? The answer lives in the **Friedmann equation** — which, underneath the intimidating notation, is really just an energy-balance statement (a cosmological cousin of the virial theorem) applied to the entire universe at once:

$$H^2 \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}.$$

Dividing through by H^2 and defining density parameters for each component relative to the **critical density** (Section 4.3) — Ω_m for matter, Ω_r for radiation, Ω_Λ for dark energy, and Ω_k for spatial curvature — this becomes the tidy normalized statement

$$\Omega_m + \Omega_r + \Omega_\Lambda + \Omega_k = 1.$$

Each component dilutes differently as the universe expands, since each responds differently to the same growing volume:

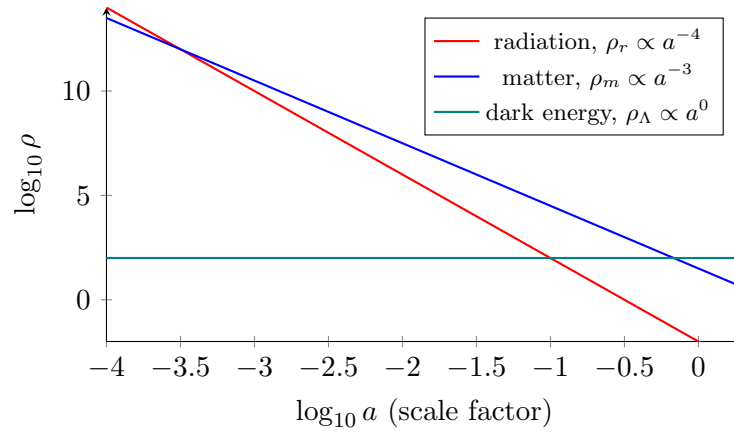


Figure 6: Energy density of each cosmic component vs. scale factor (log-log). Radiation dilutes fastest, then matter, while dark energy’s density never changes — guaranteeing it eventually dominates as the universe keeps expanding.

Matter simply dilutes as its volume grows, $\rho_m \propto a^{-3}$ — spread the same particles over a bigger box, and of course they get thinner. Radiation dilutes by that same volume factor *plus* an extra a^{-1} from the cosmological redshifting of each photon’s own energy (Section 3.3, and, well, you), giving $\rho_r \propto a^{-4}$. Dark energy is the troublemaker: its density, remarkably, does not dilute at all, $\rho_\Lambda \propto a^0$ — consistent with its strange interpretation as an intrinsic property of space itself, rather than a diluting substance sitting *within* space. Since radiation and matter both fade away while dark energy stubbornly refuses to, every dark-energy-dominated universe is doomed (or destined, depending on your outlook) to expand acceleratingly forever.

5.3 Critical Density

Somewhere between a universe that expands forever and one that eventually collapses back on itself sits a perfectly balanced knife-edge case: a spatially flat universe ($k = 0$). Setting $k = 0$ (and, for a moment, $\Lambda = 0$ for simplicity) in the Friedmann equation and solving for the exact density required to sit right on that edge defines the **critical density**:

$$H^2 = \frac{8\pi G}{3} \rho_c \quad \Rightarrow \quad \boxed{\rho_c = \frac{3H^2}{8\pi G}}$$

Current measurements place the universe’s total density remarkably close to ρ_c (i.e. $\Omega_{total} \equiv \Omega_m + \Omega_r + \Omega_\Lambda \approx 1$, so $\Omega_k \approx 0$) — strong observational support for a spatially flat universe, and one of the key motivating puzzles behind cosmic inflation, which naturally drives Ω_k toward zero regardless of its starting value.

5.4 Gravitational Lensing

Mass doesn’t just pull things toward it — it bends the very fabric the universe is drawn on, and light has no choice but to follow that bent fabric. You’d want to see just how far a purely Newtonian estimate gets here, and exactly where it falls short: treat a photon as a mass m moving at speed c on a hyperbolic flyby past a mass M (Handout 4, Section 6.1), and the small-angle deflection of that trajectory works out to $\alpha_{Newton} \sim GM/(bc^2)$, where b is the impact parameter — gravity alone, treated exactly like an ordinary gravity-assist encounter, already predicts *some* bending.

The full general-relativistic treatment (accounting for the curvature of spacetime itself, not just a force acting on a moving mass) gives exactly twice this naive estimate:

$$\alpha = \frac{4GM}{c^2 b}.$$

The extra factor of 2 is literally just a relativistic effect with no Newtonian analogue: half of the total deflection comes from light being accelerated toward the mass (the part gravity alone predicts), and the other half from space itself being curved by that same mass, bending the light's path independently of any force acting on it. This factor of 2 was, famously, the specific quantitative prediction that Eddington's 1919 solar eclipse expedition confirmed, providing one of general relativity's first great observational triumphs.

When a distant source, a massive lensing galaxy (or cluster), and the observer are all *perfectly* aligned, the symmetry of the situation bends the source's light into a complete ring around the lens — an **Einstein ring**.

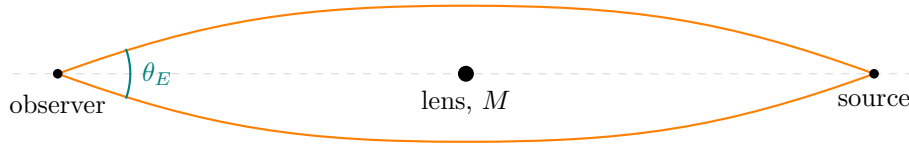


Figure 7: Perfect alignment bends light from a background source around a foreground lens symmetrically in every direction, producing a complete Einstein ring of angular radius θ_E .

Using the small-angle lens geometry (relating the deflection angle to the observer-lens, lens-source, and observer-source angular diameter distances D_L , D_{LS} , and D_S), the angular radius of this ring works out to

$$\theta_E = \sqrt{\frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S}}.$$

Since θ_E depends on the lensing mass M but on essentially nothing else about the lens's own light output, measuring an Einstein ring's angular size directly weighs the lensing object — including all of its dark matter — making gravitational lensing one of the very few mass-measurement techniques in astronomy that works even when the mass in question emits no light at all.

6 Summary Sheet

Concept	Formula
Light-gathering power	$\text{LGP} \propto D^2$
Focal ratio	$f/\# = f/D$
Rayleigh criterion	$\theta \approx 1.22\lambda/D$
Plate scale	$206265''/f \text{ (mm)}$
CCD signal-to-noise	$\text{SNR} = \frac{N_\star t}{\sqrt{N_\star t + n_{\text{pix}}(N_{\text{sky}}t + N_{\text{dark}}t + R^2)}}$
Interferometric resolution	$\theta_{\min} \sim \lambda/B_{\max}$
Hubble's Law	$v = H_0 d$
Cosmological redshift	$1 + z = a_{\text{now}}/a_{\text{then}}$
Friedmann equation	$H^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}$
Density parameter closure	$\Omega_m + \Omega_r + \Omega_\Lambda + \Omega_k = 1$
Critical density	$\rho_c = 3H^2/(8\pi G)$
Einstein ring radius	$\theta_E = \sqrt{\frac{4GM}{c^2} \frac{D_{LS}}{D_L D_S}}$